

A. Leakage-aware organ shift

Training side only

Kidney

Breast

Lung

Skin

Rank genes on training data

freeze the 50-gene panel

LOOO: unseen organ

Liver never enters training,
panel ranking, or model selection

POOLED: unseen patients

organs are known; patients
remain strictly disjoint

Two regimes answer different questions

B. Same intervention

H&E features

frozen UNI

Morphology

global slide mean
+ local neighbor mean

FiLM / adaLN-Zero

$h' = h \odot (1 + \gamma) + \beta$
re-inject morphology per layer

Matched on/off test across backbones

ST-Net

DeepSpaCE

HisToGene

Hist2ST

MLP probe

STFlow

Same folds • features • seeds • training budget

C. What COAST reveals

LOOO Δ Pearson (best FiLM variant)

ST-Net

+0.038

DeepSpaCE

+0.015

HisToGene

+0.014

MLP probe

+0.006

Hist2ST

+0.006

Gene-DML

+0.007

Main finding

FiLM improves all 6 main backbones;
full screening is in the appendix.